

Question Answering & Generative Language Models

Text and Web Mining (IS6751)

School of Communication and Information

Motivation: Question answering

- With massive collections of full-text documents, i.e., the web, simply returning relevant documents is of limited use.
- Rather, we often want **answers** to our **questions**.
- We can factor this into two parts:
 1. Finding documents that (might) contain an answer.
 - Which can be handled by traditional information retrieval/web search.
 2. Finding an answer in a paragraph or a document.
 - This problem is often termed **Reading Comprehension**.
 - It is what we will focus on in the first part of this lecture.

Question Answering

Where is the Louvre Museum located?[Sign in](#)

[Web](#) [Maps](#) [Images](#) [News](#) [Videos](#) [More ▾](#) [Search tools](#)

About 2,180,000 results (0.93 seconds)

75001 Paris, France

Louvre Museum, Address

[Feedback](#)

Louvre Museum
www.louvre.fr/

 75001 Paris, France
+33 1 40 20 50 50

Louvre - Wikipedia, the free encyclopedia
<https://en.wikipedia.org/wiki/Louvre> ▾

Jump to [Location](#), [access](#), and [facilities](#) - The Louvre or the Louvre Museum (French: Musée du Louvre, pronounced: [myze dy luvʁ]) (French (help · info)) is one of the world's largest museums and a historic monument in **Paris, France**. A central landmark of the city, it is located on the Right Bank of the Seine in the 1st arrondissement (ward).

[Louvre Palace](#) - [Louvre Pyramid](#) - [Venus de Milo](#) - [Louvre – Rivoli](#)



[See photos](#) [See outside](#)

Louvre Museum

[Directions](#)

4.6 ★★★★★ 1,809 Google reviews

\$\$ · Museum

Former historic palace housing a huge art collection, from Roman sculptures to Da Vinci's Mona Lisa.

Address: 75001 Paris, France

Phone: +33 1 40 20 50 50

Hours: Closed today · [Hours ▾](#)

Popular times 

[Tuesdays ▾](#)

Technical note: This is a “featured snippet” answer extracted from a web page, not a question answered using the (structured) Google Knowledge Graph (formerly known as Freebase).

Machine Comprehension (Burges 2013)

- “A machine **comprehends** a passage of **text** if, for any **question** regarding that text that can be **answered** correctly by a majority of native speakers, that machine can provide a **string** which those speakers would agree both answers that question and does not contain information irrelevant to that question.”

Towards the Machine Comprehension of Text: An Essay

Christopher J.C. Burges
Microsoft Research
One Microsoft Way
Redmond, WA 98052, USA

December 23, 2013



MCTest Reading Comprehension

Passage (P) + Question (Q) → Answer (A)

P

Alyssa got to the beach after a long trip. She's from Charlotte. She traveled from Atlanta. She's now in Miami. She went to Miami to visit some friends. But she wanted some time to herself at the beach, so she went there first. After going swimming and laying out, she went to her friend Ellen's house. Ellen greeted Alyssa and they both had some lemonade to drink. Alyssa called her friends Kristin and Rachel to meet at Ellen's house.....

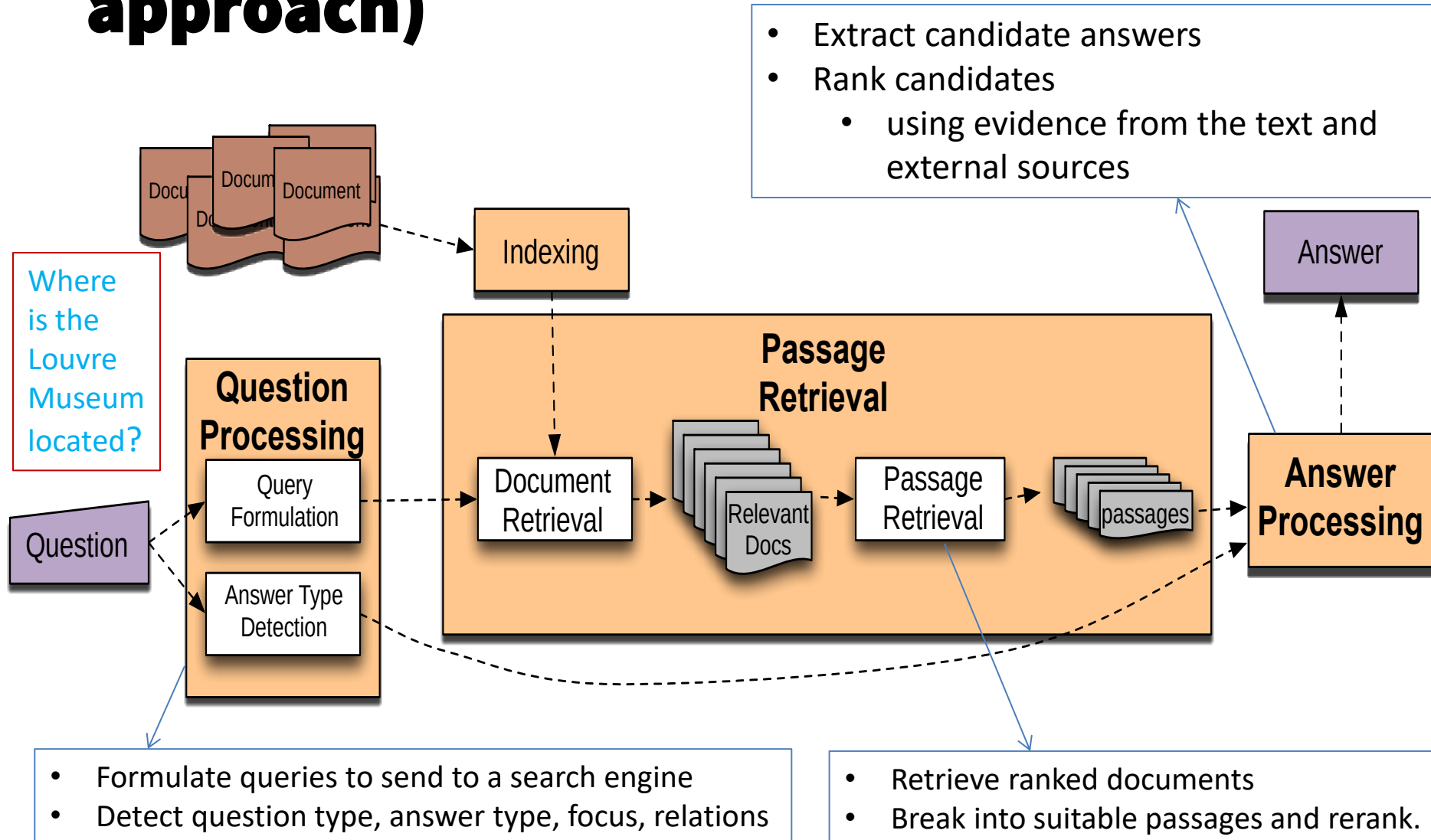
Q

Why did Alyssa go to Miami?

A

To visit some friends

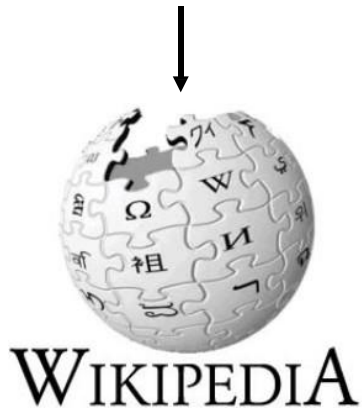
IR-based Factoid QA (traditional approach)



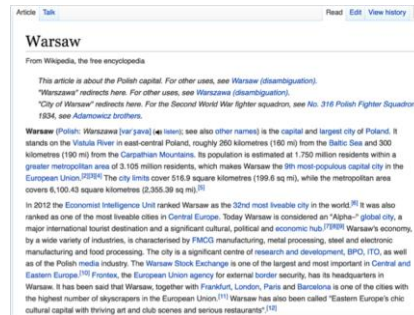
DrQA: Open-domain Question Answering using Deep Learning

(Chen, et al. ACL 2017) <https://arxiv.org/abs/1704.00051>

Q: How many of Warsaw's inhabitants spoke Polish in 1933?

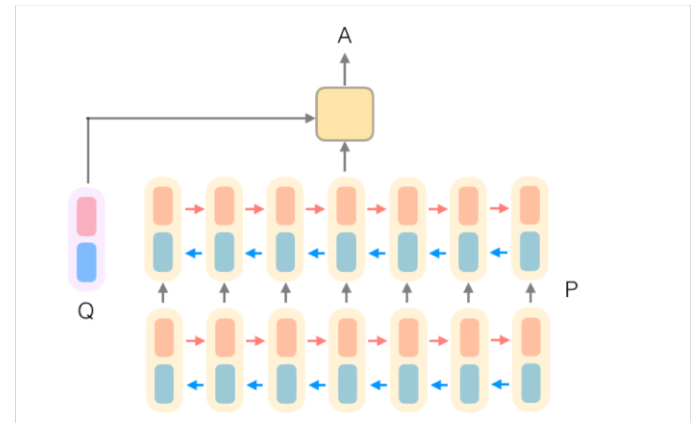


Document
Retriever



Document
Reader

833,500



Stanford Question Answering Dataset (SQuAD)

(Rajpurkar et al., 2016)

Question: Which team won Super Bowl 50?

Passage

Super Bowl 50 was an American football game to determine the champion of the National Football League (NFL) for the 2015 season. The American Football Conference (AFC) champion **Denver Broncos** defeated the National Football Conference (NFC) champion Carolina Panthers 24–10 to earn their third Super Bowl title. The game was played on February 7, 2016, at Levi's Stadium in the San Francisco Bay Area at Santa Clara, California.

100k examples

Answer must be **a span** in the passage.

A.k.a. extractive question answering

Stanford Question Answering Dataset (SQuAD v1.1)

Private schools, also known as independent schools, non-governmental, or nonstate schools, are not administered by local, state or national governments; thus, they retain the right to select their students and are funded in whole or in part by charging their students tuition, rather than relying on mandatory taxation through public (government) funding; at some private schools students may be able to get a scholarship, which makes the cost cheaper, depending on a talent the student may have (e.g. sport scholarship, art scholarship, academic scholarship), financial need, or tax credit scholarships that might be available.

Authors collected 3 gold answers for each question:

Along with non-governmental and nonstate schools, what is another name for private schools?

Gold answers: (1) independent (2) independent schools (3) independent schools

Along with sport and art, what is a type of talent scholarship?

Gold answers: (1) academic (2) academic (3) academic

Rather than taxation, what are private schools largely funded by?

Gold answers: (1) tuition (2) charging their students tuition (3) tuition

SQuAD evaluation, v1.1

- Systems are scored on two metrics:
 - **Exact match(EM)**: 1/0 accuracy on whether you match one of the 3 answers
 - **F1**: Take system and each gold answer as bag of words, evaluate.
$$\text{Precision} = \frac{TP}{TP+FP}, \text{ Recall} = \frac{TP}{TP+FN}, \text{ harmonic mean F1} = \frac{2PR}{P+R}$$

Score is average of per-question F1 scores.
- F1 measure is seen as more reliable and taken as primary.
 - It's less based on choosing exactly the same span that humans chose, which is susceptible to various effects, including line breaks.
- Both metrics ignore punctuation and articles (**a, an, the** only).

E.g., Along with non-governmental and nonstate schools, what is another name for private schools?

Gold answers: (1) independent (2) independent schools (3) independent schools

System answers: as independent schools; **bag-of-words:** { independent, schools}

Precision = 2 / 3 = 0.67; Recall: 2 / 2 = 1; F1 = 2*0.67*1 / (0.67+1) = 0.8

SQuAD v1.1 leaderboard, 2019-02-07 – it's solved!

Rank	Model	EM	F1
	Human Performance <i>Stanford University</i> (Rajpurkar et al. '16)	82.304	91.221
1 Oct 05, 2018	BERT (ensemble) <i>Google AI Language</i> https://arxiv.org/abs/1810.04805	87.433	93.160
2 Oct 05, 2018	BERT (single model) <i>Google AI Language</i> https://arxiv.org/abs/1810.04805	85.083	91.835
2 Sep 09, 2018	nlnet (ensemble) <i>Microsoft Research Asia</i>	85.356	91.202
2 Sep 26, 2018	nlnet (ensemble) <i>Microsoft Research Asia</i>	85.954	91.677
3 Jul 11, 2018	QANet (ensemble) <i>Google Brain & CMU</i>	84.454	90.490
4 Jul 08, 2018	r-net (ensemble) <i>Microsoft Research Asia</i>	84.003	90.147
5 Mar 19, 2018	QANet (ensemble) <i>Google Brain & CMU</i>	83.877	89.737
5 Sep 09, 2018	nlnet (single model) <i>Microsoft Research Asia</i>	83.468	90.133

SQuAD 2.0

- A defect of SQuAD 1.0 is that all questions have an answer in the paragraph.
- Systems (implicitly) rank candidates and choose the best one.
- You don't have to judge whether a span answers the question.
- In SQuAD 2.0, 1/3 of the training questions have no answer, and about 1/2 of the dev/test questions have no answer.
 - For NoAnswer examples, NoAnswer receives a score of 1, and any other response gets 0, for both exact match and F1.

SQuAD 2.0 Example

Genghis Khan united the Mongol and Turkic tribes of the steppes and became Great Khan in 1206. He and his successors expanded the Mongol empire across Asia. Under the reign of Genghis' third son, Ögedei Khan, the Mongols destroyed the weakened Jin dynasty in 1234, conquering most of northern China. Ögedei offered his nephew Kublai a position in Xingzhou, Hebei. Kublai was unable to read Chinese but had several Han Chinese teachers attached to him since his early years by his mother Sorghaghtani. He sought the counsel of Chinese Buddhist and Confucian advisers. Möngke Khan succeeded Ögedei's son, Güyük, as Great Khan in 1251. He

When did Genghis Khan kill Great Khan?

Gold Answers: <No Answer>

Prediction: 1234 [from Microsoft nlnet]

SQuAD 2.0 leaderboard, 2021 & 2022 – it's solved!

Rank	Model	EM	F1
	Human Performance Stanford University (Rajpurkar & Jia et al. '18)	86.831	89.452
1 Jun 04, 2021	IE-Net (ensemble) RICOH_SRCB_DML	90.939	93.214
2 Feb 21, 2021	FPNet (ensemble) Ant Service Intelligence Team	90.871	93.183
3 May 16, 2021	IE-NetV2 (ensemble) RICOH_SRCB_DML	90.860	93.100
4 Apr 06, 2020	SA-Net on Albert (ensemble) QIANXIN	90.724	93.011
5 May 05, 2020	SA-Net-V2 (ensemble) QIANXIN	90.679	92.948
5 Apr 05, 2020	Retro-Reader (ensemble) Shanghai Jiao Tong University http://arxiv.org/abs/2001.09694	90.578	92.978
5 Feb 05, 2021	FPNet (ensemble) YuYang	90.600	92.899
6 Apr 18, 2021	TransNets + SFVerifier + SFEnsembler (ensemble) Senseforth AI Research https://www.senseforth.ai/	90.487	92.894
6 Dec 01, 2020	EntitySpanFocusV2 (ensemble) RICOH_SRCB_DML	90.521	92.824
6 Jul 31, 2020	ATRLP+PV (ensemble) Hithink RoyalFlush	90.442	92.877

Source:
<https://rajpurkar.github.io/SQuAD-explorer/>

SQuAD limitations

- SQuAD has a number of other key limitations too:
 - Only span-based answers (no yes/no, counting, implicit why)
 - Questions were constructed looking at the passages.
 - Not genuine information needs
 - Generally greater lexical and syntactic matching between questions and answer span than you get in real life.
 - Barely any multi-fact/sentence inference beyond coreference.
- Nevertheless, it is a well-targeted, well-structured, clean dataset.
 - It has been the most used and competed on QA dataset.
 - It has also been a useful starting point for building systems in industry (though in-domain data always really helps!).

Stanford Attentive Reader

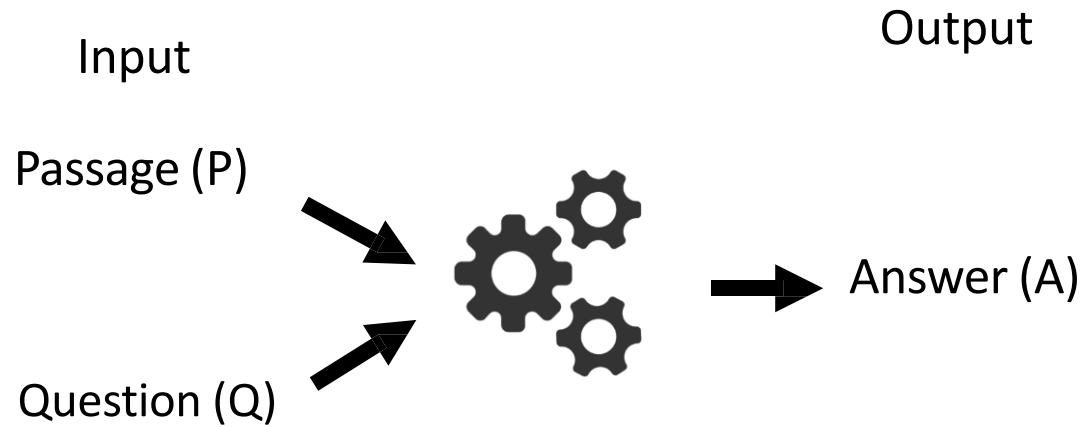
[Chen, Bolton, & Manning 2016]

[Chen, Fisch, Weston & Bordes 2017] DrQA

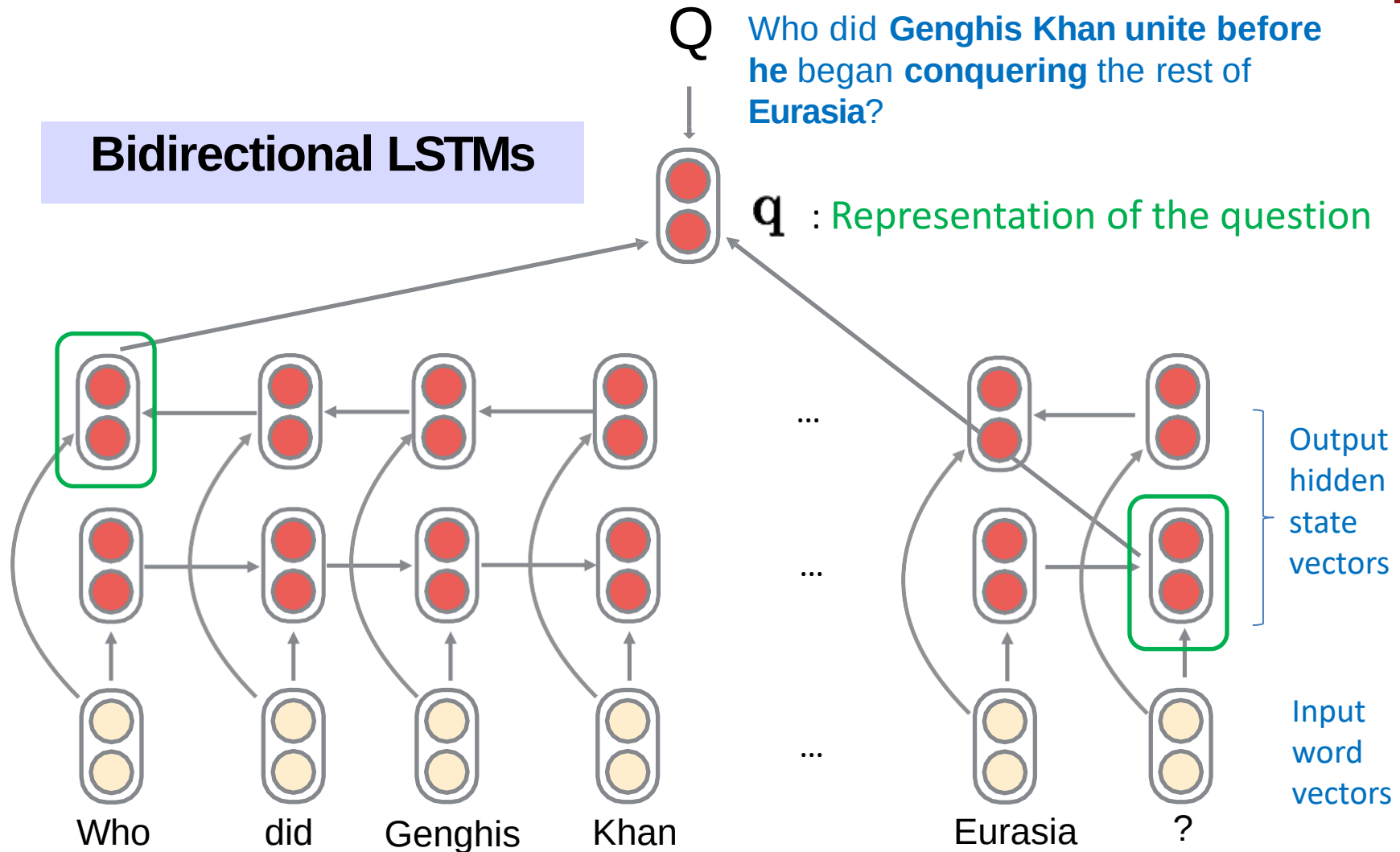
[Chen 2018]

- Demonstrated a minimal, highly successful architecture for reading comprehension and question answering.
- Became known as the Stanford Attentive Reader.

The Stanford Attentive Reader



The Stanford Attentive Reader



Stanford Attentive Reader

Q

Who did **Genghis Khan** **unite** before he began **conquering** the rest of **Eurasia**?

Bidirectional LSTMs



q

Passage(P)

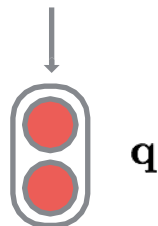
He came to power by **uniting** many of the nomadic tribes of Northeast Asia. **After** founding the Mongol Empire and being proclaimed "**Genghis Khan**", he started the Mongol invasions that resulted in the **conquest** of most of **Eurasia**. These included raids or invasions of the Qara Khitai, Caucasus, Khwarezm Empire, Western Xia and Jin dynasties. These campaigns were often accompanied by wholesale massacres of the civilian populations – especially in the Khwarezmian and Xia controlled lands. By the end of his life, the Mongol Empire occupied a substantial portion of Central Asia and China.

Answer

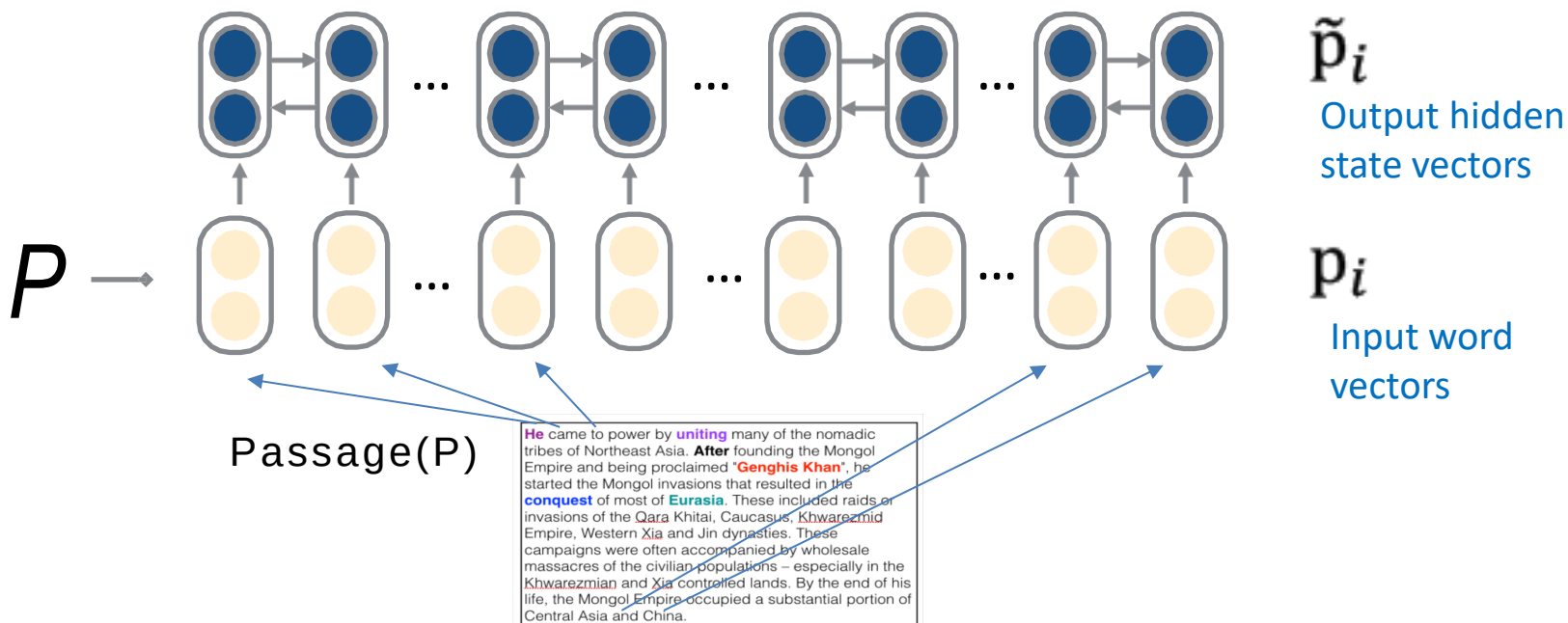


Stanford Attentive Reader

Q Who did **Genghis Khan** unite before he began **conquering** the rest of **Eurasia**?



Bidirectional LSTMs



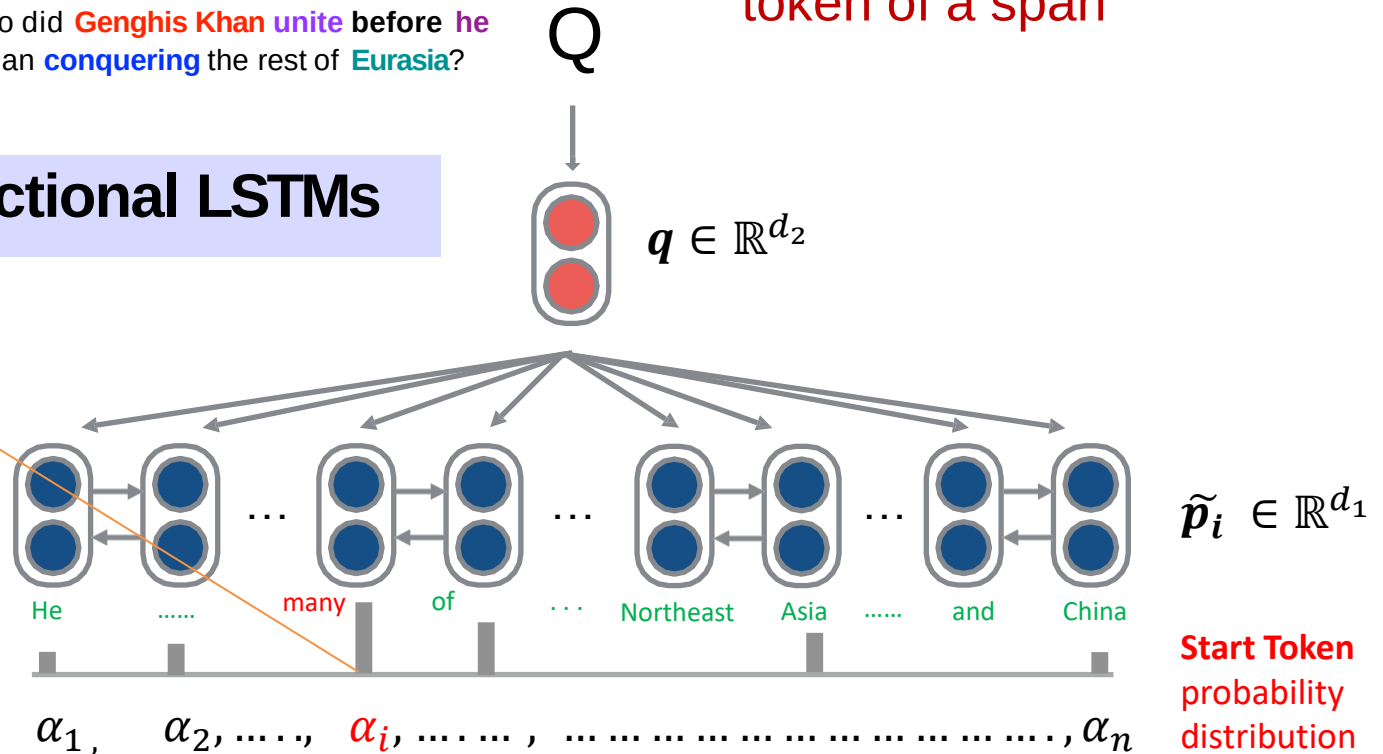
Stanford Attentive Reader

Goal: Predict the **start** token of a span

Who did **Genghis Khan** unite before he began **conquering** the rest of **Eurasia**?

Bidirectional LSTMs

He came to power by **uniting** many of the nomadic tribes of Northeast Asia. **After** founding the Mongol Empire and being proclaimed "**Genghis Khan**", he started the Mongol invasions that resulted in the **conquest** of most of **Eurasia**. These included raids or invasions of the **Qara Khitai**, Caucasus, **Khwarezm** Empire, Western **Xia** and Jin dynasties. These campaigns were often accompanied by wholesale massacres of the civilian populations – especially in the **Khwarezmian** and **Xia** controlled lands. By the end of his life, the Mongol Empire occupied a substantial portion of Central Asia and China.



Attention

Max Value

$$\alpha_i = \text{softmax}_i (q^\top W_s \tilde{p}_i)$$

→ predict **start** token

```
self.fc1 = nn.Linear (d1, d2, bias=False)
 $W_s \tilde{p}_i \leftarrow \text{self.fc1} (\tilde{p}_i) \quad \# \in \mathbb{R}^{d_2}$ 
```

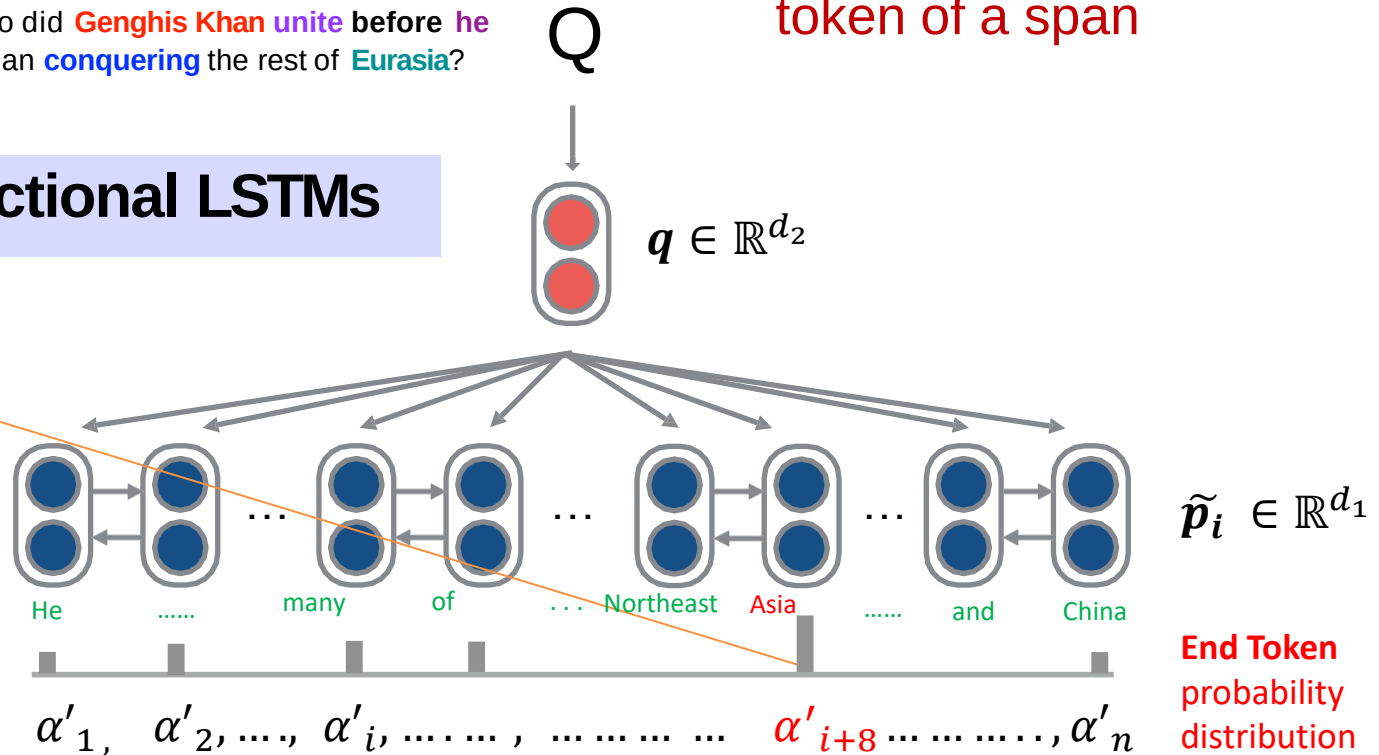
Stanford Attentive Reader

Who did **Genghis Khan** unite before he began **conquering** the rest of **Eurasia**?

Goal: Predict the **end** token of a span

Bidirectional LSTMs

He came to power by **uniting** many of the nomadic tribes of Northeast Asia. **After** founding the Mongol Empire and being proclaimed "**Genghis Khan**", he started the Mongol invasions that resulted in the **conquest** of most of **Eurasia**. These included raids or invasions of the **Qara Khitai**, Caucasus, **Khwarezm** Empire, Western **Xia** and Jin dynasties. These campaigns were often accompanied by wholesale massacres of the civilian populations – especially in the **Khwarezmian** and **Xia** controlled lands. By the end of his life, the Mongol Empire occupied a substantial portion of Central Asia and China.



Max Value

Attention

```
self.fc2 = nn.Linear (d1, d2, bias=False)
 $W'_s \tilde{p}_i \leftarrow \text{self.fc2} (\tilde{p}_i) \quad \# \in \mathbb{R}^{d_2}$ 
```

$$\alpha'_i = \text{softmax}_i (q^\top W'_s \tilde{p}_i)$$

→ predict **end** token

Stanford Attentive Reader

Training objective:

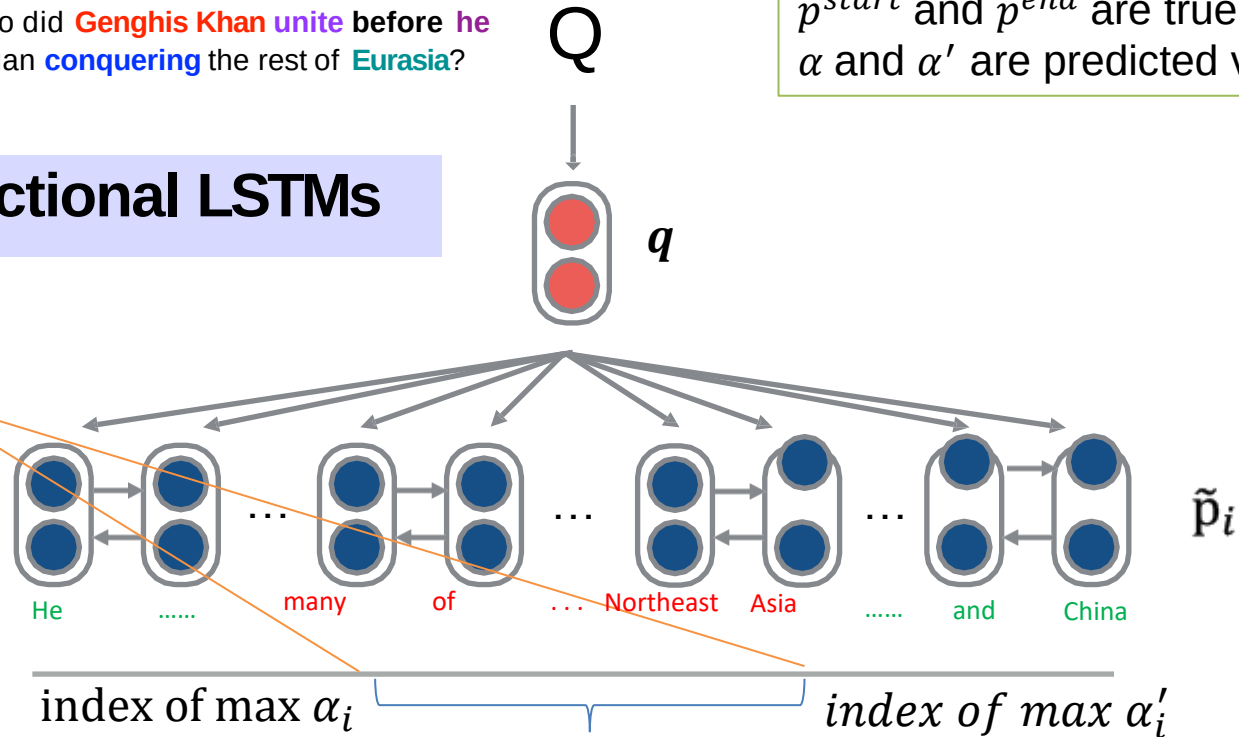
$$\mathcal{L} = -\sum_{i=1}^n p_i^{start} \log \alpha_i - \sum_{i=1}^n p_i^{end} \log \alpha'_i$$

Who did **Genghis Khan** unite before he began **conquering** the rest of **Eurasia**?

p^{start} and p^{end} are true labels.
 α and α' are predicted values.

Bidirectional LSTMs

He came to power by **uniting** many of the nomadic tribes of Northeast Asia. After founding the Mongol Empire and being proclaimed "**Genghis Khan**", he started the Mongol invasions that resulted in the **conquest** of most of **Eurasia**. These included raids or invasions of the Qara Khitai, Caucasus, Khwarezmia Empire, Western Xia and Jin dynasties. These campaigns were often accompanied by wholesale massacres of the civilian populations – especially in the **Khwarezmian** and **Xia** controlled lands. By the end of his life, the Mongol Empire occupied a substantial portion of Central Asia and China.



Example:
when true start
position is 3.

$$p^{start} = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ \dots \\ 0 \end{bmatrix} \quad \alpha = \begin{bmatrix} 0.1 \\ 0.0 \\ 0.9 \\ 0.0 \\ 0.0 \\ 0.0 \\ \dots \\ 0.0 \end{bmatrix}$$

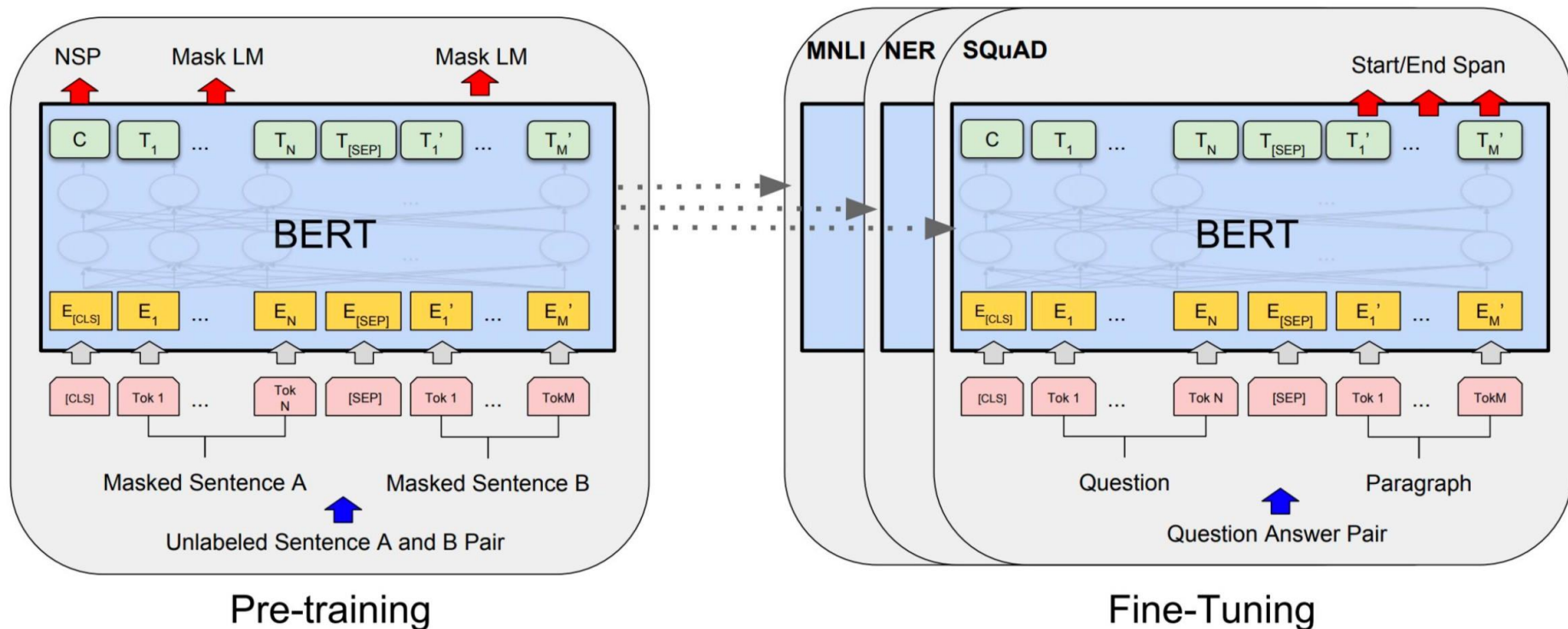
when true
end position
is 6.

$$p^{end} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ \dots \\ 0 \end{bmatrix} \quad \alpha' = \begin{bmatrix} 0.0 \\ 0.0 \\ 0.0 \\ 0.0 \\ 0.0 \\ 0.1 \\ 0.8 \\ \dots \\ 0.1 \end{bmatrix}$$

Loss =
 $-\log(0.9) - \log(0.8)$
 = 0.143

BERT model fine tuning

- Fine-tuning BERT model for Question Answering on SQuAD dataset.
 - Outperforms Stanford Attentive Model.



Next Sentence Prediction (NSP)

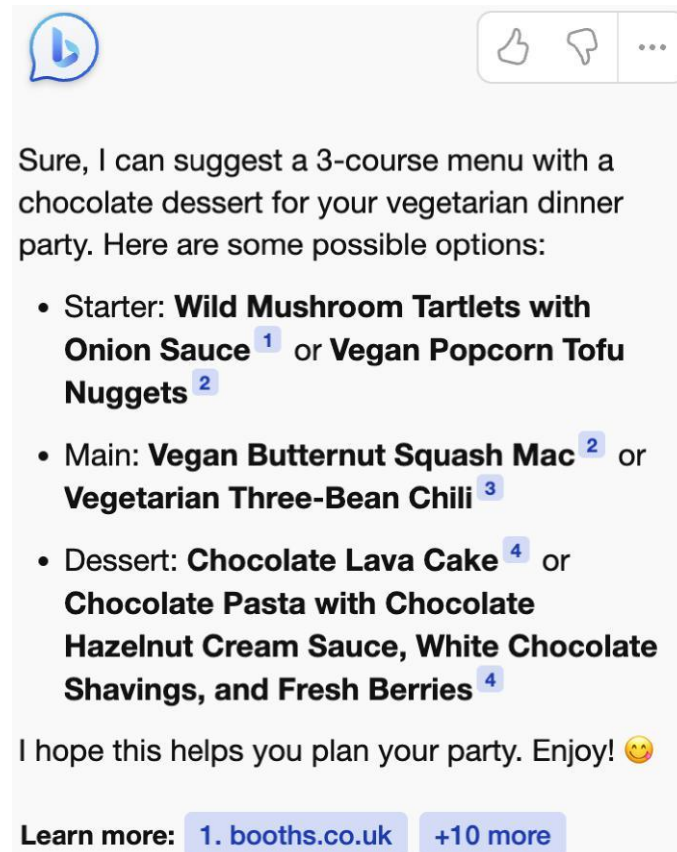
Given a pair of two sentences, the task is to say whether the second follows the first (binary classification).

Generative Language Models: Language models as multitask assistants?

I need to throw a dinner party for 6 people who are vegetarian. Can you suggest a 3-course menu with a chocolate dessert?

[Microsoft Bing]

(Also see OpenAI's ChatGPT, Google's Gemini, Anthropic's Claude)



The screenshot shows a chat window with the Microsoft Bing logo in the top left and interaction icons (thumbs up, thumbs down, and a menu) in the top right. The chat text reads: "Sure, I can suggest a 3-course menu with a chocolate dessert for your vegetarian dinner party. Here are some possible options:" followed by a bulleted list of menu items. Each item has a small blue square with a number (1, 2, 3, or 4) next to it, indicating a link. The list includes: Starter: Wild Mushroom Tartlets with Onion Sauce¹ or Vegan Popcorn Tofu Nuggets²; Main: Vegan Butternut Squash Mac² or Vegetarian Three-Bean Chili³; Dessert: Chocolate Lava Cake⁴ or Chocolate Pasta with Chocolate Hazelnut Cream Sauce, White Chocolate Shavings, and Fresh Berries⁴. Below the list, it says "I hope this helps you plan your party. Enjoy! 😊". At the bottom, there is a "Learn more:" section with a link "1. booths.co.uk" and a "+10 more" button.

Sure, I can suggest a 3-course menu with a chocolate dessert for your vegetarian dinner party. Here are some possible options:

- Starter: **Wild Mushroom Tartlets with Onion Sauce**¹ or **Vegan Popcorn Tofu Nuggets**²
- Main: **Vegan Butternut Squash Mac**² or **Vegetarian Three-Bean Chili**³
- Dessert: **Chocolate Lava Cake**⁴ or **Chocolate Pasta with Chocolate Hazelnut Cream Sauce, White Chocolate Shavings, and Fresh Berries**⁴

I hope this helps you plan your party. Enjoy! 😊

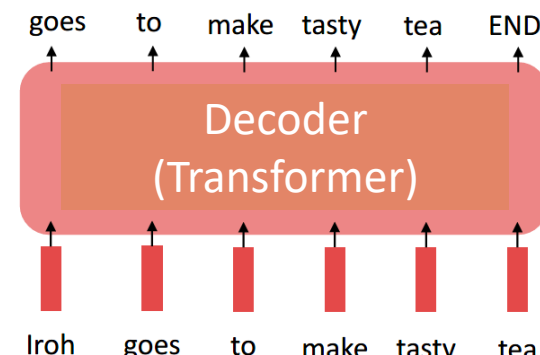
Learn more: [1. booths.co.uk](#) [+10 more](#)

Emergent abilities of large language models: GPT (2018)

Let's look at the **Generative Pretrained Transformer (GPT)** models from OpenAI as an example:

GPT (117M parameters; Radford et al., 2018)

- Transformer decoder with 12 layers.
- Trained on BooksCorpus: over 7000 unique books (4.6GB text)



GPT showed that language modeling at scale can be an effective pretraining technique for downstream tasks like **natural language inference**.

entailment

[START] *The man is in the doorway* [DELIM] *The person is near the door* [EXTRACT]

Target categories: entailment/contradictory/neutral

Emergent abilities of large language models: GPT-2 (2019)

GPT-2 (1.5B parameters; Radford et al., 2019)

- Same architecture as GPT, just bigger (117M -> **1.5B**)
 - But trained on **much more data**: 4GB -> **40GB** of Internet text data (WebText)
 - Scrape links posted on Reddit with at least 3 upvotes (rough proxy of human quality)

Language Models are Unsupervised Multitask Learners

Alec Radford *¹ Jeffrey Wu *¹ Rewon Child¹ David Luan¹ Dario Amodei **¹ Ilya Sutskever **¹

Note: BERT Large has **340M** parameters.

Emergent **zero-shot** learning

One key emergent ability in GPT-2 is **zero-shot learning**: the ability to do many tasks with **no examples**, and **no gradient updates**, by simply:

- Specifying the right sequence prediction problem (e.g., question answering):

Passage: Tom Brady... Q: Where was Tom Brady born? A: ...

- GPT-2 beats SoTA (State-of-The-Art) on language modeling benchmarks with no task-specific fine-tuning.

Emergent abilities of large language models: GPT-3 (2020)

GPT-3 (175B parameters; Brown et al., 2020)

- Another increase in parameters size (1.5B -> **175B**)
- and data (40GB -> **over 600GB**)

Language Models are Few-Shot Learners

Tom B. Brown*

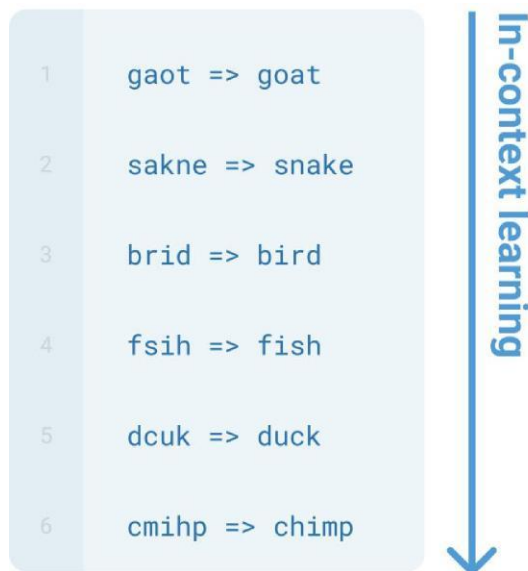
Benjamin Mann*

Nick Ryder*

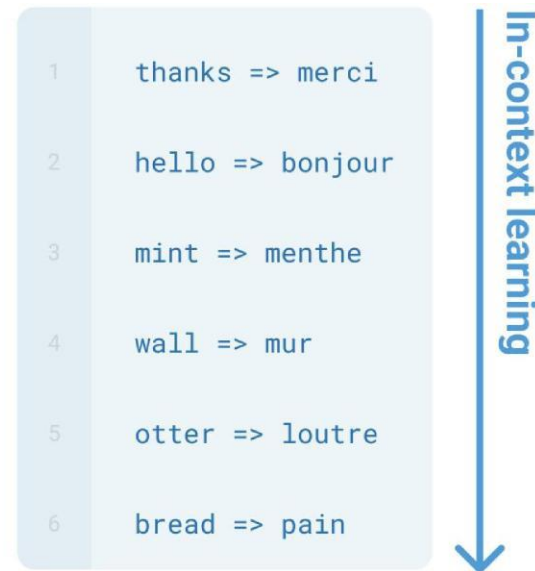
Melanie Subbiah*

Emergent **few-shot** learning

- Specify a task by simply **prepending examples of the task before your example**
- Also called **in-context learning**, to stress that *no gradient updates* are performed when learning a new task.



anelysis =>



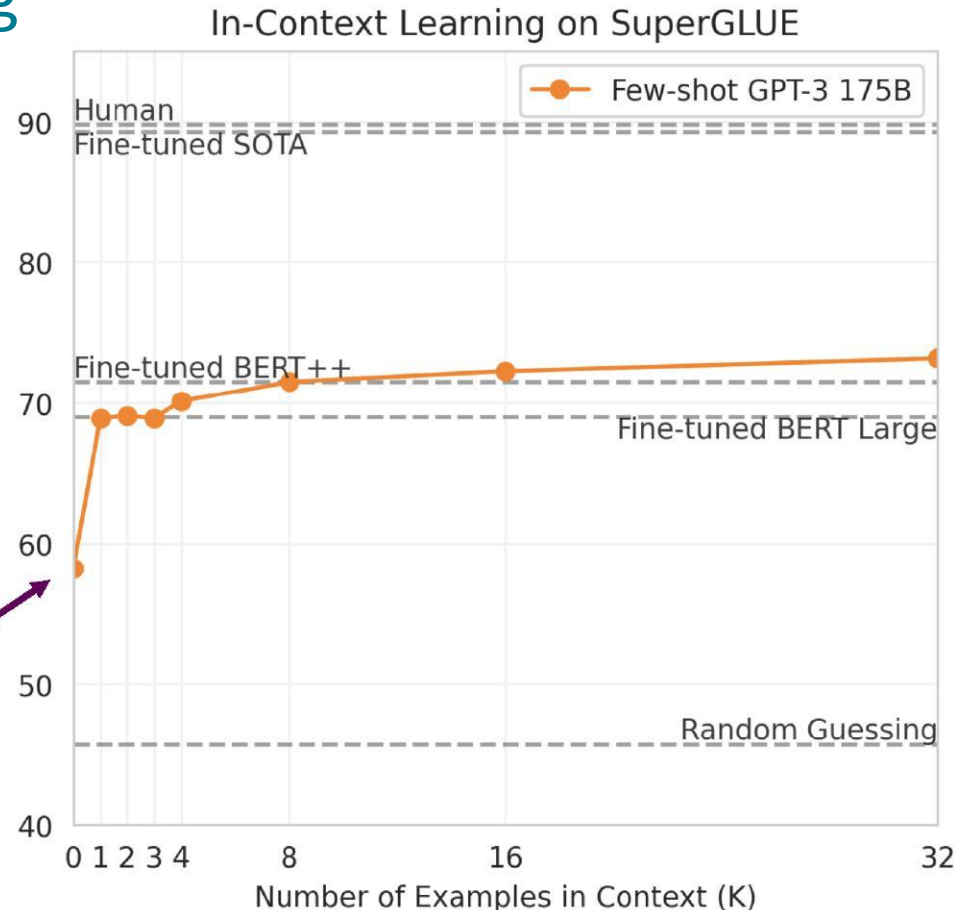
flower =>

[Brown et al., 2020]

Emergent few-shot learning

Zero-shot

- 1 Translate English to French:
- 2 cheese =>



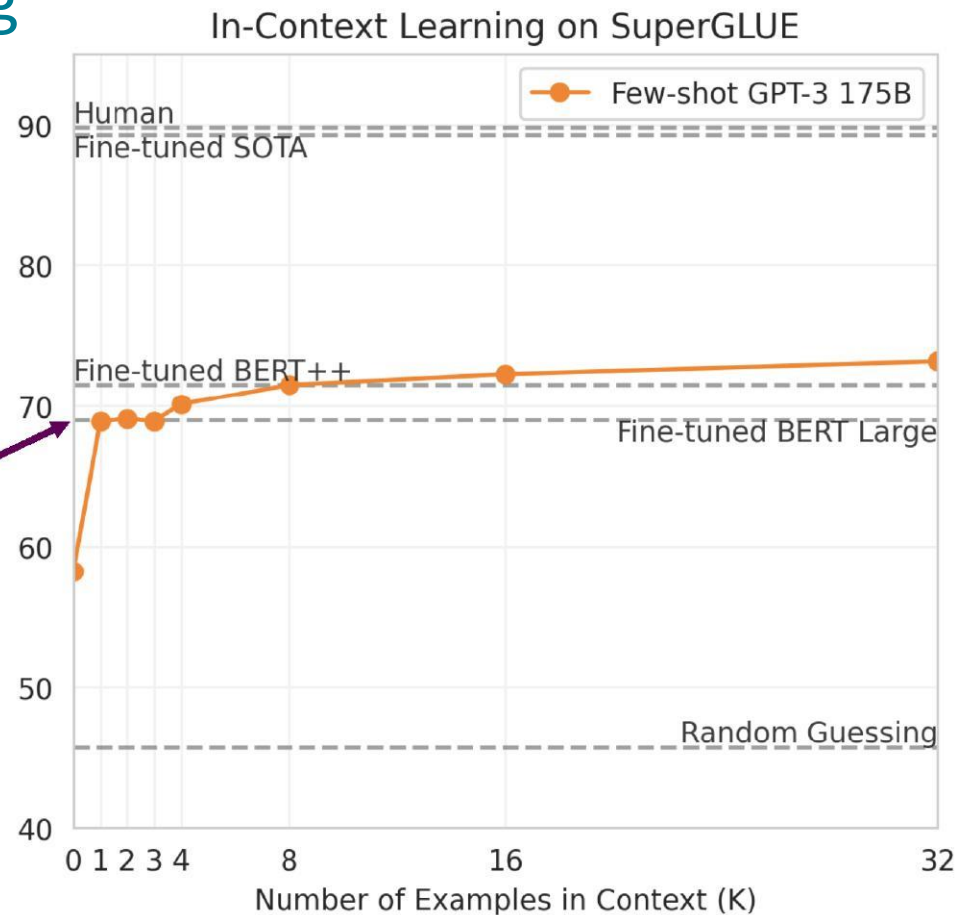
[Brown et al., 2020]

Note: SuperGLUE, a new benchmark styled after GLUE with a new set of more difficult language understanding tasks.

Emergent few-shot learning

One-shot

1 Translate English to French:
2 sea otter => loutre de mer
3 cheese =>

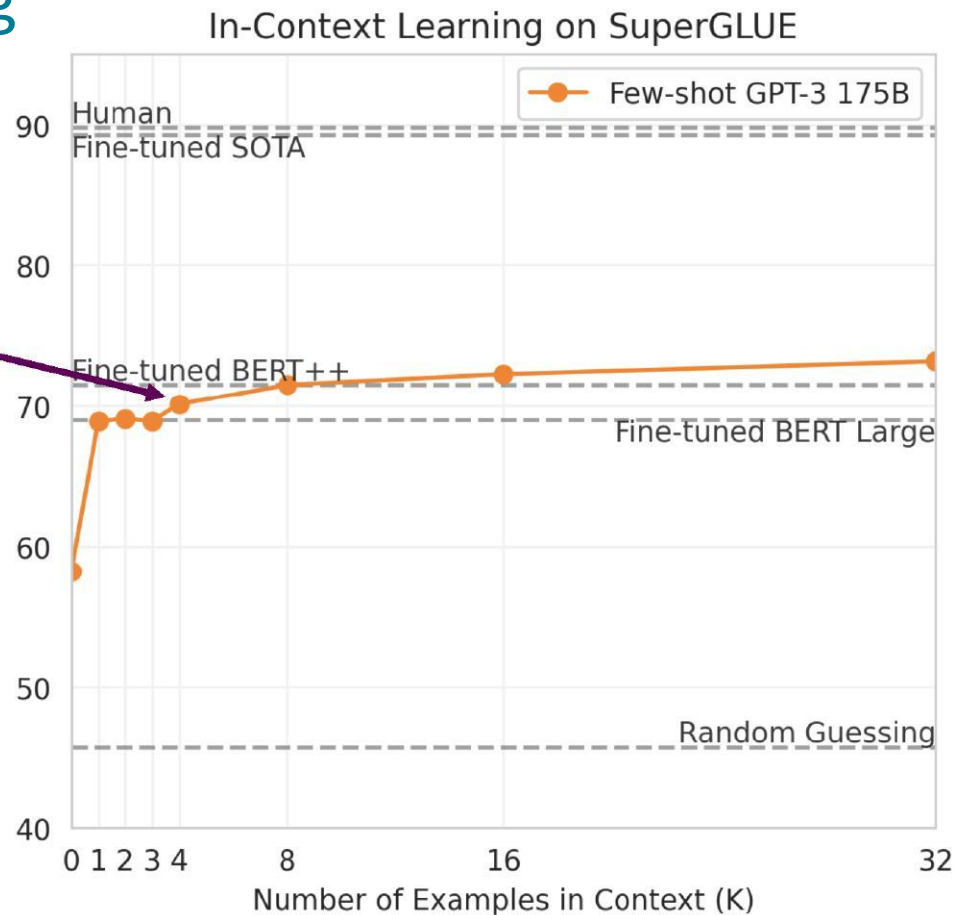


[Brown et al., 2020]

Emergent few-shot learning

Few-shot

1 Translate English to French:
2 sea otter => loutre de mer
3 peppermint => menthe poivrée
4 plush girafe => girafe peluche
5 cheese =>



[Brown et al., 2020]

Chain-of-thought prompting

Standard Prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The answer is 27. ❌

Chain-of-Thought Prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. $5 + 6 = 11$. The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The cafeteria had 23 apples originally. They used 20 to make lunch. So they had $23 - 20 = 3$. They bought 6 more apples, so they have $3 + 6 = 9$. The answer is 9. ✅

[Wei et al., 2022; also see Nye et al., 2021]

Chain-of-thought prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

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Model Output

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Do we even need
examples of reasoning?
Can we just ask the model
to reason through things?

[Wei et al., 2022; also see Nye et al., 2021]

Zero-shot chain-of-thought prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. $5 + 6 = 11$. The answer is 11.

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Model Output

A: The cafeteria had 23 apples originally. They used 20 to make lunch. So they had $23 - 20 = 3$. They bought 6 more apples, so they have $3 + 6 = 9$. The answer is 9. ✓

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A: **Let's think step by step.** There are 16 balls in total. Half of the balls are golf balls. That means there are 8 golf balls. Half of the golf balls are blue. That means there are 4 blue golf balls. ✓

[Kojima et al., 2022]

From Language Models to Assistants

1. Zero-Shot (ZS) and Few-Shot (FS) In-Context Learning

- + No finetuning needed, prompt engineering (e.g., Chain of Thought) can improve performance
- Limits to what you can fit in context
- Complex tasks will probably need gradient steps

2. Instruction finetuning

3. Reinforcement Learning from Human Feedback (RLHF)

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Language modeling $\neq$ assisting users

PROMPT *Explain the moon landing to a 6 year old in a few sentences.*

COMPLETION GPT-3

Explain the theory of gravity to a 6 year old.

Explain the theory of relativity to a 6 year old in a few sentences.

Explain the big bang theory to a 6 year old.

Explain evolution to a 6 year old.

Language models are not *aligned* with user intent
[Ouyang et al., 2022].

Language modeling $\neq$ assisting users

PROMPT

Explain the moon landing to a 6 year old in a few sentences.

COMPLETION

Human

A giant rocket ship blasted off from Earth carrying astronauts to the moon. The astronauts landed their spaceship on the moon and walked around exploring the lunar surface. Then they returned safely back to Earth, bringing home moon rocks to show everyone.

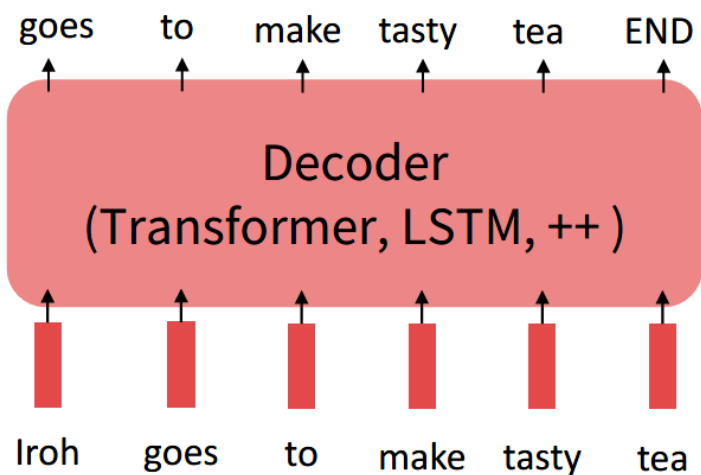
- Language models are not *aligned* with user intent [Ouyang et al., 2022].
- **Finetuning to the rescue!**

Recall from previous Lecture: The Pretraining / Finetuning Paradigm

Pretraining can improve NLP applications by serving as parameter initialization

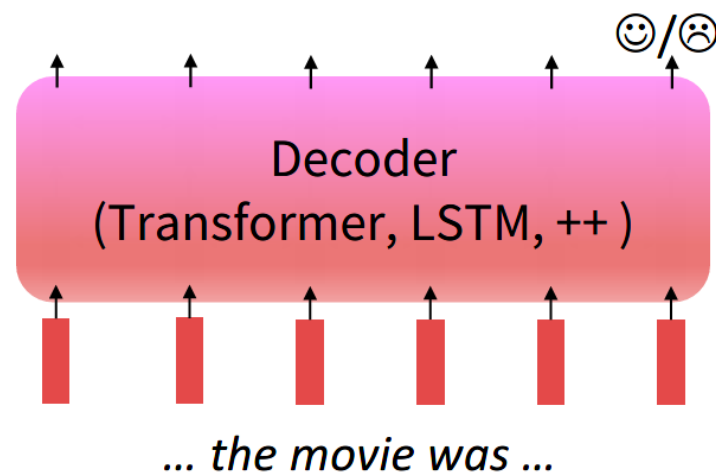
Step 1: Pretrain (on language modeling)

Lots of text; learn general things!



Step 2: Finetune (on your task)

Not many labels; adapt to the task!

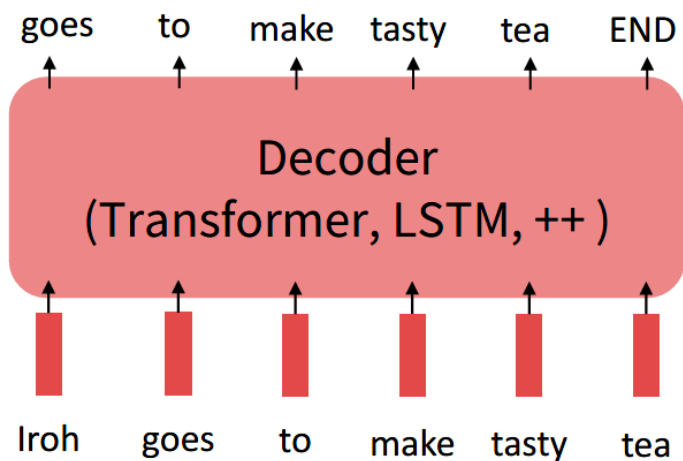


Scaling up finetuning

Pretraining can improve NLP applications by serving as parameter initialization

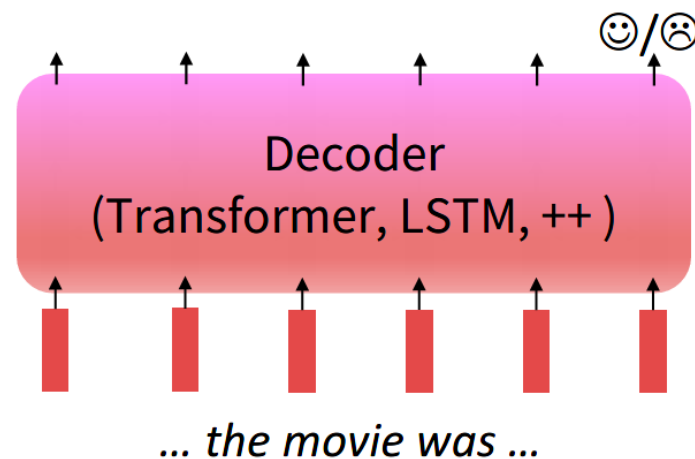
Step 1: Pretrain (on language modeling)

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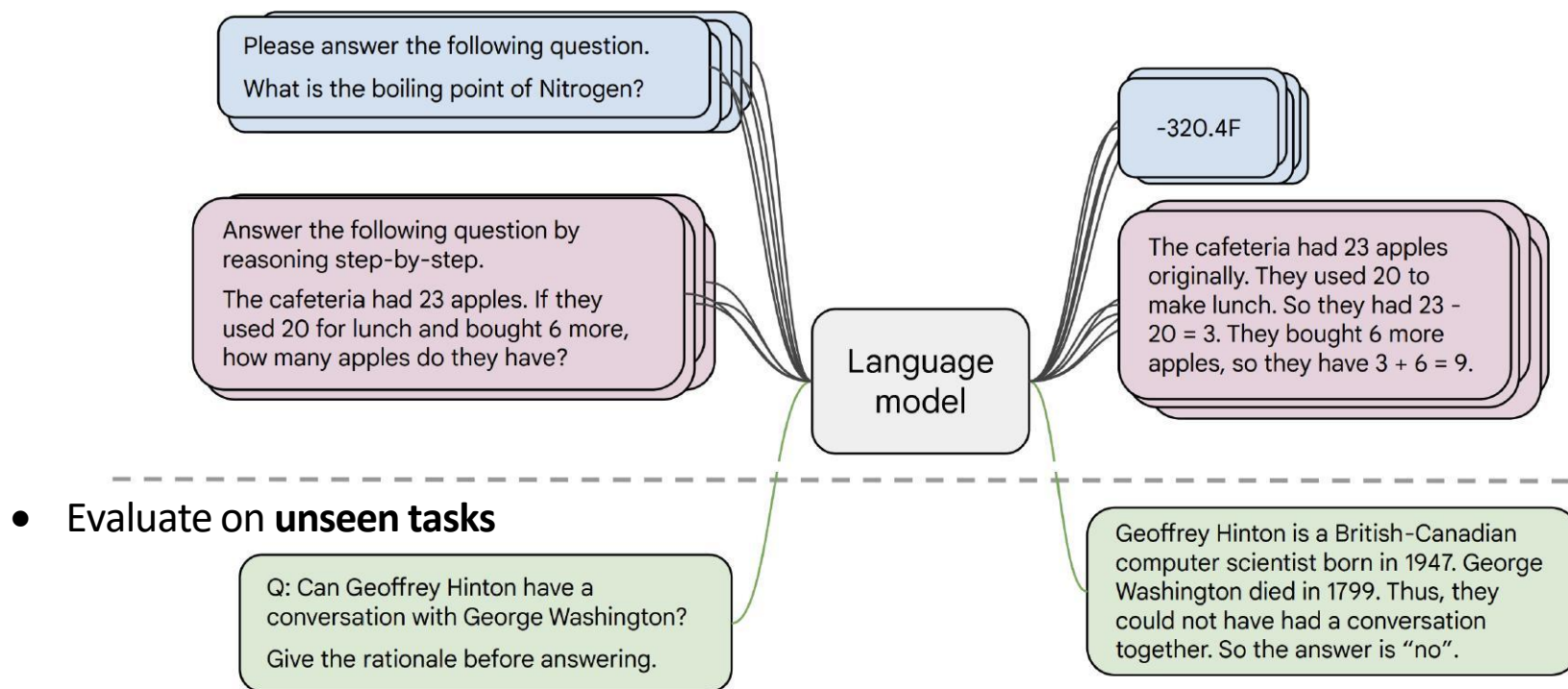
Step 2: Finetune (on **many tasks**)

~~Not~~ many labels; adapt to the tasks!



Instruction finetuning

Collect examples of (instruction, output) pairs across many tasks and finetune an LM



[FLAN-T5; [Chung et al., 2022](#)]

Instruction tuning *typically* uses the *same loss function* (usually cross-entropy loss) across various tasks — because it treats all tasks as "next-token prediction" given the instruction.

Instruction ~~finetuning~~ pretraining?

- As is usually the case, **data + model scale** is key for this to work!
- For example, the **Super-NaturalInstructions** dataset contains **over 1.6K tasks**, **3M+** examples
 - Classification, sequence tagging, rewriting, translation, QA...
- **Q:** how do we evaluate such a model after instruction finetuning?



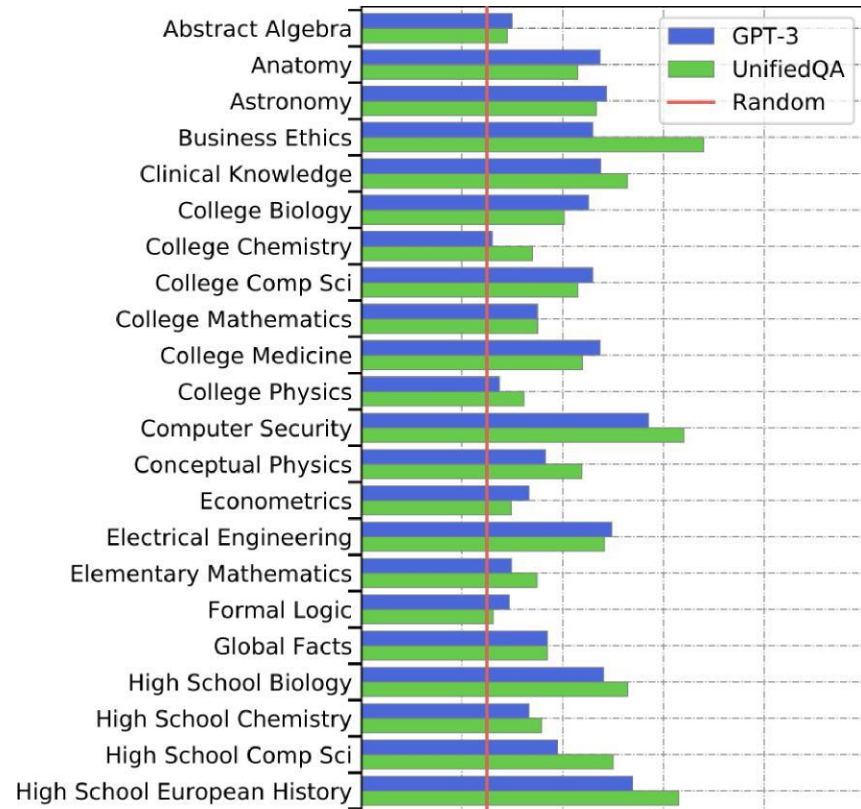
New benchmarks for multitask LMs

Massive Multitask Language Understanding (MMLU)

[Hendrycks et al., 2021]

- New benchmarks for measuring LM performance on **57** diverse *knowledge intensive* tasks.
- **Multiple-Choice questions** using few-shot in-context learning.

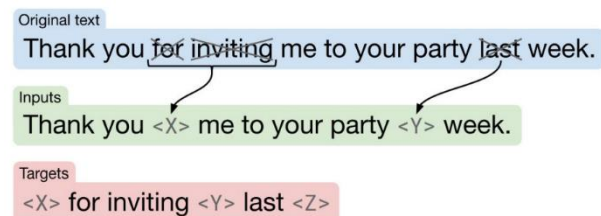
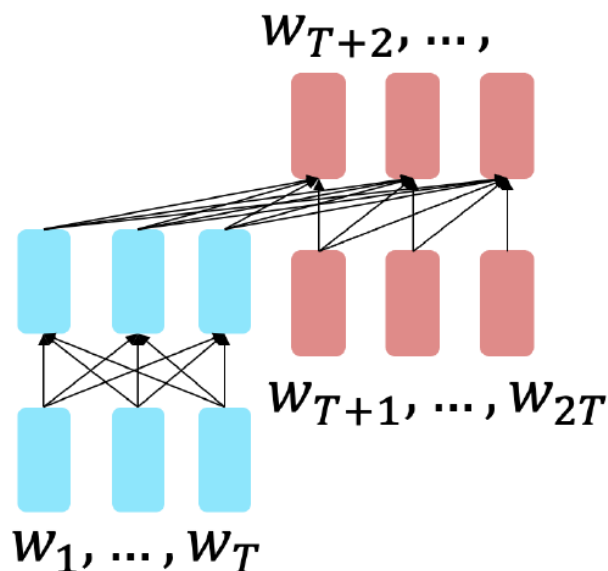
<https://github.com/hendrycks/test>



[illegible]

Instruction finetuning

- The **T5** encoder-decoder model [Raffel et al., 2018] pretrained on the **span corruption** task.
- Flan-T5** [Chung et al., 2020]: **T5** models finetuned on 1.8K additional tasks.



Params	Model	BIG-bench + MMLU avg (normalized)
80M	T5-Small	-9.2
	Flan-T5-Small	-3.1 (+6.1)
250M	T5-Base	-5.1
	Flan-T5-Base	6.5 (+11.6)
780M	T5-Large	-5.0
	Flan-T5-Large	13.8 (+18.8)
3B	T5-XL	-4.1
	Flan-T5-XL	19.1 (+23.2)
11B	T5-XXL	-2.9
	Flan-T5-XXL	23.7 (+26.6)

Bigger model = bigger Δ

[Chung et al., 2022]

Instruction finetuning

Model input (Disambiguation QA)

Q: In the following sentences, explain the antecedent of the pronoun (which thing the pronoun refers to), or state that it is ambiguous.

Sentence: The reporter and the chef will discuss their favorite dishes.

Options:

- (A) They will discuss the reporter's favorite dishes
- (B) They will discuss the chef's favorite dishes
- (C) Ambiguous

A: Let's think step by step.

Before instruction finetuning

The reporter and the chef will discuss their favorite dishes.

The reporter and the chef will discuss the reporter's favorite dishes.

The reporter and the chef will discuss the chef's favorite dishes.

The reporter and the chef will discuss the reporter's and the chef's favorite dishes.

✗ (doesn't answer question)

Highly recommend trying FLAN-T5 out to get a sense of its capabilities:

<https://huggingface.co/google/flan-t5-xxl> [Chung et al., 2022]

Instruction finetuning

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A: Let's think step by step.

After instruction finetuning

The reporter and the chef will discuss their favorite dishes does not indicate whose favorite dishes they will discuss. So, the answer is (C). ✓

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<https://huggingface.co/google/flan-t5-xxl>

[[Chung et al., 2022](#)]

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2. Instruction finetuning

- + Simple and straightforward, generalize to unseen tasks
-
-

3. Reinforcement Learning from Human Feedback (RLHF)

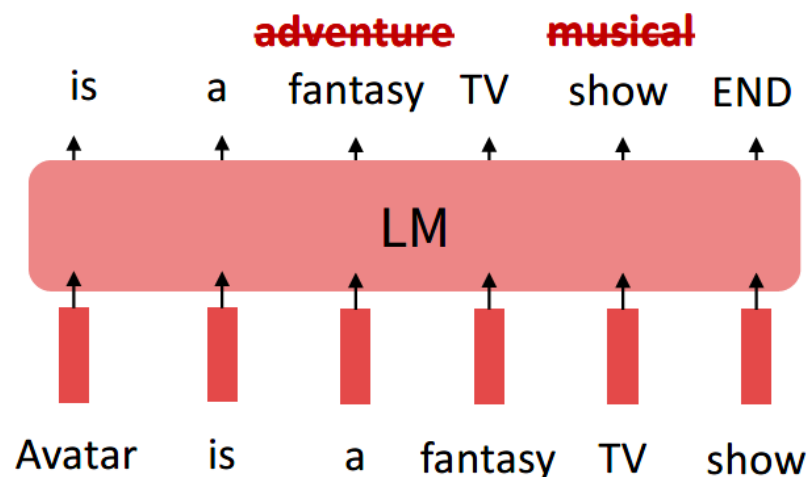
Limitations of instruction finetuning?

- One limitation of instruction finetuning is obvious: it's **expensive** to collect ground-truth data for tasks.
- But there are other, subtler limitations too.
- **Problem 1:** tasks like open-ended creative generation have no right answer.

- *Write me a story about a dog and her pet grasshopper.*

- **Problem 2:** language modeling penalizes all token-level mistakes equally, but some errors are worse than others.

- Even with instruction finetuning, there a mismatch between the LM objective and the objective of “satisfy human preferences”!
- Can we **explicitly attempt to satisfy human preferences**?



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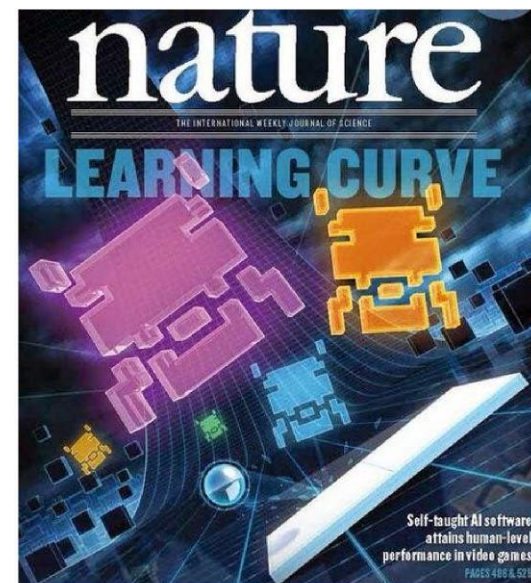
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3. Reinforcement Learning from Human Feedback (RLHF)

Reinforcement learning to the rescue

- The field of **reinforcement learning (RL)** has studied these (and related) problems for many years now [Williams, 1992; Sutton and Barto, 1998]
- In 2013, resurgence of interest in RL applied to deep learning, game-playing [Mnih et al., 2013]
- But the interest in applying RL to modern LMs is an even newer phenomenon [Ziegler et al., 2019; Stiennon et al., 2020; Ouyang et al., 2022]. **Why?**
 - RL with LMs has commonly been viewed as very hard to get right (still is!)
 - Newer advances in RL algorithms that work for large neural models, including language models (e.g., **PPO (Proximal Policy Optimization)**); [Schulman et al., 2017])



InstructGPT: scaling up RLHF to tens of thousands of tasks

30k
tasks!

Step 1

**Collect demonstration data,
and train a supervised policy.**

A prompt is
sampled from our
prompt dataset.

Explain the moon
landing to a 6 year old

A labeler
demonstrates the
desired output
behavior.

Some people went
to the moon...

This data is used
to fine-tune GPT-3
with supervised
learning.

SFT

(Instruction finetuning)

Step 2

**Collect comparison data,
and train a reward model.**

A prompt and
several model
outputs are
sampled.

Explain the moon
landing to a 6 year old

A Explain gravity...
B Explain war...
C Moon is natural
satellite of...
D People went to
the moon...

A labeler ranks
the outputs from
best to worst.

D > C > A = B

This data is used
to train our
reward model.

RM
D > C > A = B

(Reward Model)

Step 3

**Optimize a policy against
the reward model using
reinforcement learning.**

A new prompt
is sampled from
the dataset.

Write a story
about frogs

The policy
generates an
output.

PPO

Once upon a time...

The reward model
calculates a
reward for
the output.

RM

The reward is
used to update
the policy
using PPO.

r_k

[Ouyang et al., 2022]

(RLHF)

InstructGPT: scaling up RLHF to tens of thousands of tasks

Tasks collected from labelers:

- **Plain:** We simply ask the labelers to come up with an arbitrary task, while ensuring the tasks had sufficient diversity.
- **Few-shot:** We ask the labelers to come up with an instruction, and multiple query/response pairs for that instruction.
- **User-based:** We had a number of use-cases stated in waitlist applications to the OpenAI API. We asked labelers to come up with prompts corresponding to these use cases.

Use-case	Prompt
Brainstorming	List five ideas for how to regain enthusiasm for my career
Generation	Write a short story where a bear goes to the beach, makes friends with a seal, and then returns home.

[\[Ouyang et al., 2022\]](#)

InstructGPT

PROMPT *Write a short poem about a wise frog.*

COMPLETION GPT-3

Write a short story in which a character has two different names.

Write a short story in which you try to get something back that you have lost.

Write a short story in which a character has a bad dream.

InstructGPT

The frog is a symbol of wisdom
He knows all the secrets of the world
He is a master of disguise
And a great teacher of life
He is a symbol of transformation
And the bringer of change
He is the frog who has seen it all
And knows the meaning of it all

ChatGPT: Instruction Finetuning + RLHF for dialog agents

ChatGPT: Optimizing Language Models for Dialogue

Note: OpenAI (and similar companies) are keeping more details secret about ChatGPT training (including data, training parameters, model size)—perhaps to keep a competitive edge...

Methods

We trained this model using Reinforcement Learning from Human Feedback (RLHF), using the same methods as InstructGPT, but with slight differences in the data collection setup. We trained an initial model using supervised fine-tuning: human AI trainers provided conversations in which they played both sides—the user and an AI assistant. We gave the trainers access to model-written suggestions to help them compose their responses. We mixed this new dialogue dataset with the InstructGPT dataset, which we transformed into a dialogue format.

(Instruction finetuning!)

<https://openai.com/blog/chatgpt/>

ChatGPT: Instruction Finetuning + RLHF for dialog agents

ChatGPT: Optimizing Language Models for Dialogue

Methods

To create a reward model for reinforcement learning, we needed to collect comparison data, which consisted of two or more model responses ranked by quality. To collect this data, we took conversations that AI trainers had with the chatbot. We randomly selected a model-written message, sampled several alternative completions, and had AI trainers rank them. Using these reward models, we can fine-tune the model using Proximal Policy Optimization. We performed several iterations of this process.

(RLHF!)

<https://openai.com/blog/chatgpt/>

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3. Reinforcement Learning from Human Feedback (RLHF)

- + Directly model preferences (cf. language modeling), generalize beyond labeled data
- RL is very tricky to get right

Limitations of RL + Reward Modeling

- Human preferences are unreliable!
 - Chatbots are rewarded to produce responses that *seem* authoritative and helpful, *regardless of truth*.
 - This can result in making up facts + hallucinations.
- **Models** of human preferences are *even more* unreliable!
- There is a real concern of AI mis(alignment)!



Percy Liang
@percylang

...

RL from human feedback seems to be the main tool for alignment. Given reward hacking and the falliability of humans, this strategy seems bound to produce agents that merely appear to be aligned, but are bad/wrong in subtle, inconspicuous ways. Is anyone else worried about this?

10:55 PM · Dec 6, 2022

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- RL is very tricky to get right
- Human preferences are fallible; *models* of human preferences even more so

Referenced Materials

- Natural Language Processing with Deep Learning,
<http://web.stanford.edu/class/cs224n/>
- NLP, Dan Jurafsky and Christopher Manning,
<http://www.stanford.edu/~jurafsky/NLPCourseraSlides.html>